

ASHWIN M DEVANGA

PhD in Computer Science, University of North Dakota, Grand Forks, ND

MS in Data Analytics Engineering, Northeastern University, Boston, MA

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INTERESTS

- Reinforcement Learning
- Machine Learning and Data Mining
- Autonomous Systems and Path Planning
- Quantum Circuits: Quantum Machine Learning

EDUCATION

Level of Education	Major / University	Institute	Year	Grades
Graduate Specialization: Graduation(PhD)	<i>Computer Science</i> Univ. of North Dakota	Univ. of North Dakota, Grand Forks, ND	2025- current	3.83/4
Graduate Specialization: Graduation(M.S)	<i>Data Analytics Engineering</i> Northeastern University	Northeastern University, Boston, MA	2020- 2022	3.79/4
UnderGraduate Specialization: UnderGraduation(B.Tech)	<i>Biotechnology Engineering</i> IIT Guwahati	IIT Guwahati, Guwahati, India	2016- 2020	6.01/10

TECHNICAL SKILLS

- **Languages** (Python, R, MATLAB, JAVA, C)
- **Scripting** (Linux shell [bash, zsh])
- **Tools** (GIT, Make, Tableau, Simio, $\LaTeX$)

EXPERIENCE

- **Graduate Research and Teaching Assistant** (SEECs, University of North Dakota) (*Jan'25 - Current*)
 - Supervisor: Dr. Tingjun Lei
 - Courses Taught: Secure Software Engineering, Object-Oriented Programming, Digital Circuits Lab
- **Software Engineer in Test - MATLAB Math, Data Analysis and SciML tools** (Quality Engineering, The Mathworks) (*Oct'22 - Oct'24*)
 - Work on Core Math and Data Analysis Functions in MATLAB.
 - Build test strategies/procedures and implement them.
 - Projects include automatic differentiation and Generative AI for MATLAB copilot - Data Analysis.
- **Teaching Assistant** (MIE-COE, Northeastern University) (*Jan'21 - Aug'22*)
 - Supervisor: Dr. Dehghanimohammadabadi Mohammad,
 - Courses Taught: Reinforcement Learning, Metaheuristics and Applications, Simulation Analysis.
- **Research Assistant** (MIE-COE, Northeastern University) (*Jan'21 - Aug'22*)
 - Reinforcement Learning and Metaheuristics for Operations Research Problems.
 - Worked on Operations Research problems such as the Vehicle Routing Problem and Job Scheduling.
 - Environment and Tools used: MATLAB, Python, C#, Simio.
- **Research Internship** (Chubu University, Japan) (*May 2018 – Jul 2018, May 2019 – Jul 2019*)
 - Supervisor: Prof. K. Yamauchi,
 - Project based on machine learning on a budget. Implementation of Supervised Reinforcement learning using Actor-Critic model for one-step learning.
 - Development of a modified actor-critic algorithm with dual actors to improve performance.
 - Environment used: Python with PyTorch and JAVA.

PUBLICATIONS

- R. Amin, I. Boukabou, M. Abdelsalam, A. Devanga, T. Lei and N. Kaabouch (2026). **Comparative Analysis of Time-Series and Non-Time-Series Deep Learning Models for UAV Trajectory Prediction Under Remote ID Packet Loss.** *2026 Integrated Communications, Navigation and Surveillance Conference (ICNS), Herndon, VA, USA, 2026* pp. 1-7.
DOI: 10.1109/ICNS69853.2026.11570520
- A. Devanga, R. Amin, M. Abdelsalam, I. Boukabou, T. Lei and N. Kaabouch (2026). **Maintaining UAV Surveillance During Remote ID Signal Loss Using Predictive Trajectory Modeling.** *2026 Integrated Communications, Navigation and Surveillance Conference (ICNS), Herndon, VA, USA, 2026* pp. 1-6.
DOI: 10.1109/ICNS69853.2026.11570577
- M. Abdelsalam, I. Boukabou, R. Amin, A. Devanga, T. Lei and N. Kaabouch (2026). **Impact of GNSS Signal Loss on UAV Trajectory Prediction Using Multimodal Deep Learning Models.** *2026 Integrated Communications, Navigation and Surveillance Conference (ICNS), Herndon, VA, USA, 2026* pp. 1-7.
DOI: 10.1109/ICNS69853.2026.11570329
- A. Devanga, T. Lei, J. Hu, C. Luo and S. Shao (2025). **Deep Reinforcement Learning for USV Navigation in Dynamic Ocean Environments.** *IEEE International Conference on Robotics and Biomimetics (IEEE ROBIO), Chengdu, China, 2025* pp. 1024-1029.
DOI: 10.1109/ROBIO66223.2025.11375987
- A. Devanga, T. Lei, J. Hu, J. Chen and C. Luo, (2025). **Trustworthy Cyber-Resilient Reinforcement Learning for Secure Navigation under Adversarial Attack.** *Cyber Awareness and Research Symposium (CARS), Grand Forks, ND, USA, 2025* pp 1-6.
DOI: 10.1109/CARS67163.2025.11337781
- Belsare, S., Devanga, A., Dehghanimohammadabadi, M. (2024). **AI-Driven Multi-Objective UAV Route Optimization.** *2024 Winter Simulation Conference (WSC), Orlando, FL, USA, 2024* pp. 2619-2630.
DOI: 10.1109/WSC63780.2024.10838909
- Devanga, A., Badilla, E.D. and Dehghanimohammadabadi, M. (2022). **Applied Reinforcement Learning for Decision Making in Industrial Simulation Environments.** *In 2022 Winter Simulation Conference (WSC)* (pp. 2819-2829).
DOI: 10.1109/WSC57314.2022.10015282.
- Devanga, A. and Yamauchi, K. (2019). **Collaborative Learning of Human and Computer: Supervised Actor-Critic based Collaboration Scheme.** *In Proceedings of the 8th International Conference on Pattern Recognition Applications and Methods - Volume 1: ICPRAM, ISBN 978-989-758-351-3, pages 794-801.*
DOI: 10.5220/0007568407940801

AWARDS

- ICNS 2026 : Best of Track Paper Award - **Impact of GNSS Signal Loss on UAV Trajectory Prediction with Multi-Modal Deep Learning Models.**
- IEEE CARS 2025 : Best Paper Award - **Trustworthy Cyber-Resilient Reinforcement Learning for Secure Navigation under Adversarial Attack.**

CONFERENCES AND TALKS

- IEEE International Conference on Robotics and Biomimetics 2025 - Speaker : **Deep Reinforcement Learning for USV Navigation in Dynamic Ocean Environments.** Chengdu, China.
- Cyber Awareness and Research Symposium 2025 - Speaker : **Trustworthy Cyber-Resilient Reinforcement Learning for Secure Navigation under Adversarial Attack.** Grand Forks, ND, USA.
- Winter Simulation Conference 2022 - Speaker : **Applied Reinforcement Learning for Decision Making in Industrial Simulation Environments.** Singapore.
- INFORMS Annual Meet 2021 - Session Chair and Speaker : **DM-Gym: A set of environments for developing reinforcement learning agents to solve Data Mining problems.** Anaheim, Los Angeles, CA, USA.
- ICPRAM 2019 - Poster Presentation : **Collaborative Learning of Human and Computer: Supervised Actor-Critic based Collaboration Scheme.** Prague, Czech Republic.